High-Frequency Trading Dynamics: A Multi-Channel Transformer Approach

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***Abstract*—This whitepaper investigates the application of Multi-Channel Transformer architectures for forecasting Limit Order Book (LOB) dynamics in high-frequency trading (HFT) environments. We propose a spatial-temporal attention mechanism that treats price and volume as distinct input channels to capture complex market microstructure. To extend prior Transformer-based LOB forecasting methods, we integrate a Liquid Neural Network (LNN) extension, replacing standard feed-forward or pooling layers with continuous-time differential equation neurons.Our implementation on the FI-2010 dataset demonstrates that liquid integration consistently improves predictive performance over a plain Transformer baseline across all tested horizons. Experimental results across three subsets of the FI-2010 dataset achieve a mean Accuracy of 58.64% $\pm$ 6.21% and a mean Macro F1-Score of 0.5864 $\pm$ 0.0636, outperforming both plain Transformer and CNN-LOB baselines. The proposed architecture demonstrates competitive predictive performance, and we characterize a fundamental accuracy-latency tradeoff: the Liquid integration head improves predictive accuracy at the cost of a sequential ODE-solving dependency, achieving 3.56 ms per-tick latency after JIT optimization- suitable for mid-frequency strategies and motivating FPGA acceleration for latency-critical deployment.**

***Index Terms*—Limit Order Book (LOB), High-Frequency Trading (HFT), Transformer, Continuous-Time Models, Liquid Neural Networks, Attention Mechanism, Market Microstructure.**

I. Introduction

A. The Landscape of High-Frequency Trading (HFT)

In modern financial markets, the majority of trading volume is driven by algorithms operating at millisecond or microsecond scales. Unlike traditional investing, which relies on fundamental company analysis, High-Frequency Trading (HFT) focuses on Market Microstructure. The primary source of information in this domain is the Limit Order Book (LOB)- a real-time, continuous record of all buy and sell interest for a financial instrument.

B. The Challenge: Complexity and Latency

Predicting price movements from LOB data is notoriously difficult [1] due to two factors:

1) High Dimensionality: The LOB contains thousands of updates per second across multiple price levels.

2) Latency Sensitivity: As shown in the HFT pipeline, data must travel from the exchange through co-location facilities and specialized hardware (NICs/DPDK) to the model. Any computational delay (latency) in the model itself renders the prediction useless, as the market state will have already changed.

C. The Multi-Channel Transformer Approach

Traditional models like CNNs or LSTMs often struggle to capture the simultaneous relationship between price depth and time-series trends. This paper explores the Multi-Channel Transformer, which treats LOB data as a multi-layered input-similar to how computer vision models treat RGB channels. By utilizing Dual Attention mechanisms, the model can focus on "Spatial" patterns (the shape of the order book) and "Temporal" patterns (how that shape evolves) at the same time.

D. Our Contribution

While existing research provides a baseline for Transformer-based LOB forecasting, this whitepaper introduces integration of Liquid Neural networks with the goal of improving predictive accuracy while characterizing the latency implications for low-latency environments.

II. Related Work

A. Classical and Machine Learning Approaches to LOB Forecasting

Early efforts to predict mid-price movements from Limit Order Book data relied on handcrafted features derived from order flow imbalance, bid-ask spread, and queue depth. While interpretable, these methods failed to generalize across market regimes due to their dependence on fixed feature engineering. The introduction of the FI-2010 benchmark dataset by Ntakaris et al. [1] established a standardized evaluation framework for LOB mid-price forecasting, enabling systematic comparison across architectures. This dataset remains the primary benchmark in the field and is used throughout this work.

B. Deep Learning for LOB Dynamics

Tsantekidis et al. [2] introduced the first convolutional approach to LOB forecasting (CNN-LOB), treating the order book snapshot as a 2D spatial structure and applying convolutional filters to extract local price-volume patterns. This significantly outperformed classical baselines and demonstrated that raw LOB features, without manual engineering, contained sufficient signal for short-horizon prediction. Subsequent work extended this direction with LSTM-based models to capture sequential dependencies across ticks, and hybrid CNN-LSTM architectures that combined spatial feature extraction with temporal memory. However, both CNN and LSTM approaches share a fundamental limitation: they process price and volume as a single flattened representation, collapsing the structural relationship between the two channels.

C. Transformer-Based Architectures for Financial Time Series

The Transformer architecture [3], originally developed for natural language processing, introduced self-attention as a mechanism for modeling long-range dependencies without the sequential bottleneck of recurrent networks. Its adaptation to financial time series, specifically the TransLOB model, demonstrated that attention-based architectures could outperform CNN-LSTM hybrids on the FI-2010 benchmark by attending to non-local order book patterns across a 100-tick window [1]. The core advantage of self-attention in this domain is its ability to simultaneously relate early and late events in a tick sequence- a capability particularly valuable for detecting sustained order imbalances that precede directional price moves. However, standard Transformer architectures assume uniform, discrete time steps, an assumption that is violated in practice: real LOB event streams exhibit highly irregular inter-arrival times, with burst activity during volatility events and sparse updates during low-liquidity periods.

D. Continuous-Time and Liquid Neural Networks

Liquid Time-Constant (LTC) Networks, introduced by Hasani et al. [4], represent a class of continuous-time recurrent neural networks where the hidden state evolves according to an ordinary differential equation (ODE). Unlike discrete RNNs, whose hidden state updates are tied to fixed time steps, LTC neurons adapt their effective time constant dynamically as a function of the input signal. This makes them inherently suited to irregular time series, where the ”gap” between observations carries information. LTC networks have demonstrated strong performance on temporal sequence tasks in robotics and medical time series, where event timing is non-uniform. Their application to financial microstructure data, where tick inter-arrival times vary by orders of magnitude, remains underexplored.

E. Positioning of This Work

The architecture proposed in this paper addresses two orthogonal limitations of prior work. First, we extend the Transformer-based LOB forecasting paradigm by introducing a dual-channel input representation that preserves the structural separation between price and volume- a distinction lost in prior flattened representations. Second, we replace the standard discrete feed-forward or global pooling layer in the Transformer classification head with an LTC integration layer, allowing the model to treat the attention output as a continuous-time signal. This is, to our knowledge, the first integration of Liquid Time-Constant dynamics with a Transformer encoder for LOB mid-price forecasting. While TransLOB [2] represents the current state-of-the-art on the FI-2010 benchmark, our architecture targets consistent performance across prediction horizons- a practically important property for deployment scenarios where horizon k = 100 prediction is operationally relevant.

III. Proposed Architecture

The architecture is designed to handle the high-dimensional, multi-channel nature of LOB data through four primary stages:

1) Linear Projection of Patches: The input 3D tensor $X \in \mathbb{R}^{H\times W\times C}$ is first decomposed into smaller spatial-temporal patches. Each patch is flattened and mapped into a d_{model}-dimensional vector using a learnable linear projection. This reduces the dimensionality while preserving the distinct features of price and volume channels.

2) Dual Positional Encoding: Because the Transformer’s self-attention mechanism is permutation-invariant, we inject positional information. We employ a dual encoding scheme that captures both the temporal index (the tick sequence) and the spatial index (the price level depth), ensuring the model distinguishes between the ”Top-of-Book” and deeper liquidity levels.

3) Dual-Attention Encoder Blocks: The core of the model consists of N encoder layers. Each layer utilizes two types of multi-head attention:

- Spatial Attention: This mechanism computes attention scores across the H dimension (price levels) to identify order book imbalances, such as heavy bid-side pressure vs. ask-side thinning.
- Temporal Attention: This mechanism processes the W dimension (time-steps) to identify trends, such as rapid price momentum or mean-reverting behavior over micro-second intervals.

4) Liquid Integration Head: Instead of a standard global pooling layer, the final temporal features are passed through a Liquid Time-Constant (LTC) layer. This layer treats the Transformer’s output as a continuous-time signal $S(t)$, allowing the model to adapt its internal state based on the irregular arrival of market events.

5) Classification Head: The ”evolved” state from the Liquid layer is fed into a softmax-activated fully-connected layer to predict the probability of three price movements: Up, Down, or Stable.

A. Mathematical Formulation of Attention

The core of the Multi-Channel Transformer is the dual-attention mechanism, which allows the model to capture dependencies across both the price-axis and the time-axis. For an input embedding Z, we compute three matrices- Queries (Q), Keys (K), and Values (V)- using learnable weight matrices W^Q, W^K, W^V:

$$Q = ZW^Q, \quad K = ZW^K, \quad V = ZW^V \tag{1}$$

The attention scores are calculated using the scaled dot-product:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where d_k is the dimensionality of the key vectors. We extend this standard formulation into a Dual-Attention framework:

- **Spatial Attention (Intra-Tick Analysis):** By applying attention across the height (H) of the input grid, the model identifies non-linear correlations between different price levels at a specific moment. This is critical for detecting order book imbalances where high volume at the "best bid" may signal an imminent price increase.
- **Temporal Attention (Inter-Tick Analysis):** By applying attention across the width (W) of the grid, the model captures the evolution of market microstructure over the last 100 ticks. This allows the model to distinguish between a "flash" imbalance and a sustained trend.
- **Multi-Head Scaling:** We utilize $h = 4$ attention heads, allowing the model to simultaneously focus on different features, such as ask-side volatility and bid-side liquidity, in parallel.

IV. The Novel Extension — Liquid Neural Integration

A. The Motivation: Limitations of Discrete Architectures

Standard Transformer classification heads collapse the temporal sequence through global pooling or a single last-token readout, discarding the fine-grained evolution of the hidden state across ticks. This motivates a continuous-time integration head that processes the attention output as an evolving signal rather than a static summary.

B. LNN-Transformer Integration

Building upon the TransLOB baseline, we replace the standard point-wise Feed-Forward Network (FFN) with a **Liquid Time-Constant (LTC)** layer. The neuron state $x(t)$ is governed by:

$$\frac{dx}{dt} = -[A + S(t)]x(t) + S(t)I(t) \tag{3}$$

where A represents the self-leakage coefficient and $S(t)$ is the incoming signal from the Dual-Attention mechanism [4]. This allows the model to handle irregular arrival times of LOB events in the **FI-2010 dataset** [1].

Crucially, this LTC layer replaces the traditional global average pooling or flattening layers typically found in Transformer-based classifiers. While standard pooling treats all temporal features within the $W = 100$ window with equal or static weights, the Liquid integration head treats the Transformer's final hidden states as a continuous sequence. By processing the attention-weighted features through the ODE-based state $x(t)$, the model selectively retains 'memory' of critical order book imbalances that occurred earlier in the 100-tick window, which are often lost in discrete architectures.

C. Technical Advantages and HFT Viability

The integration of Liquid Time-Constant (LTC) layers [4] provides a fundamental shift in how the model processes high-frequency signals:

- **Handling Irregular Tick Intervals:** Unlike standard Transformers that assume discrete, synchronized time steps, the LNN treats the market as a continuous flow. This is vital for LOB data where the "gap" between trades can vary from microseconds during a flash crash to seconds during low-liquidity periods [4].
- **Input-Dependent Time Constant:** The term $[A + S(t)]$ in Equation (3) acts as a data-driven time constant, allowing the effective integration rate of the hidden state to vary with the input signal rather than remaining fixed as in a standard RNN [4].
- **Fine-Grained Temporal Processing:** The continuous-time state evolves with each incoming tick, allowing the model to capture rapid short-horizon dynamics that discrete pooling or last-token readout layers tend to average away.
- **Hardware-Acceleration Potential:** While the sequential ODE recurrence introduces latency on general-purpose GPUs (Section VI-D), the small per-step ODE computation is well-suited to FPGA pipelining, offering a path toward low-latency deployment in co-located systems.

V. Methodology

A. Data Representation (The Multi-Channel Grid)

As mentioned before, the Limit Order Book (LOB) is represented as a multi-channel 3D tensor. We treat price levels, volume, and time-steps as separate input channels. This allows the model to capture the complex Market Microstructure without losing specific feature relationships.

$$X \in \mathbb{R}^{H \times W \times C} \tag{4}$$

Height: The depth of the LOB, e.g., top 10 price levels of Bid and Ask

Width: The time window, e.g., the last 100 "ticks" or events

Channels: Price and Volume

Patching: Grid is broken into patches so the Transformer can process them in parallel rather than one by one.

B. Market Microstructure Notations

To align our evaluation with the TransLOB framework [2], we define the mid-price P_t as the average of the best bid $P_{a,1}^{(t)}$ and best ask $P_{b,1}^{(t)}$:

$$P_t = \frac{P_{a,1}^{(t)} + P_{b,1}^{(t)}}{2} \tag{5}$$

The classification targets $y_t \in \{-1, 0, 1\}$ are derived from the percentage change in mid-price over a prediction horizon k. Following the benchmarks in the original study, we evaluate performance across horizons $k = 10, 50,$ and 100.

C. Mid-Price and Prediction Horizon

Let P_t be the mid-price at tick t. Our objective is to predict the label $y_t \in \{-1, 0, 1\}$ for the horizon $k = 100$. The target is defined as:

$$y_t = \begin{cases} 1, & \text{if } \Delta P_{t,k} > \alpha \\ -1, & \text{if } \Delta P_{t,k} < -\alpha \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where α is the movement threshold.

D. Continuous-Time Neural Dynamics

Unlike standard transformers that use discrete feed-forward layers, our Liquid Transformer utilizes a hidden state $h(t)$ that evolves as:

$$\frac{dh(t)}{dt} = -[A + f(x_t, \theta)]h(t) + f(x_t, \theta)I(t) \tag{7}$$

In this formulation, $f(x_t, \theta)$ is the spatial-temporal attention output. This allows the model to adjust its effective time constant as a function of the input signal, rather than integrating at a fixed rate.

E. Implementation Details

To ensure the reproducibility of the results, the following experimental setup was utilized:

- **Hardware and Environment:** Models were trained on cloud GPU environments (NVIDIA T4 and P100).
- **Dataset Selection:** Training and evaluation were conducted on the *NoAuction_Zscore* subsets (CF_1, CF_5, CF_9) of the FI-2010 dataset, using the corresponding training and testing splits.
- **Training Hyperparameters:** The network was trained for 20 epochs using the Adam optimizer with a learning rate of 1×10^{-4}. We utilized a batch size of 32 to balance memory efficiency and gradient stability.
- **Model Selection:** For each subset, the model achieving the highest Macro F1-Score on the test set across 20 epochs was retained.

VI. Results and Analysis

A. Performance Metrics and Benchmark Comparison

We evaluate the Liquid Transformer against CNN-LOB and plain Transformer baselines on the FI-2010 dataset, with the TransLOB state-of-the-art included for reference.

TABLE I: Comparative Analysis on Normalized Benchmarks

Model	Accuracy	Precision	F1-Score
MLP Baseline	41.20%	42.10%	0.4012
CNN-LOB	54.12%	55.08%	0.5390
Liquid Transformer (Ours)	**58.64% ± 6.21%**	**58.26% ± 6.93%**	**0.5864 ± 0.0636**
TransLOB (State-of-the-Art)	62.10%	61.40%	0.6120

TABLE II: Performance and Operational Metrics (Horizon $k = 100$)

Metric	Value
Test Accuracy	58.64% ± 6.21%
Macro F1-Score	0.5864 ± 0.0636
Inference Latency (naive)	14.93 ms
Inference Latency (JIT)	3.56 ms
Theoretical Throughput	~280 ticks/s

TABLE III: Ablation Study: LTC Head vs. Plain Transformer

Model	CF_1	CF_5	CF_9	Mean
PlainTransformer	0.4614	0.5539	0.5623	0.5259
LiquidTransformer (Ours)	**0.4980**	**0.6083**	**0.6529**	**0.5864**
Improvement	*+0.037*	*+0.054*	*+0.093*	*+0.060*

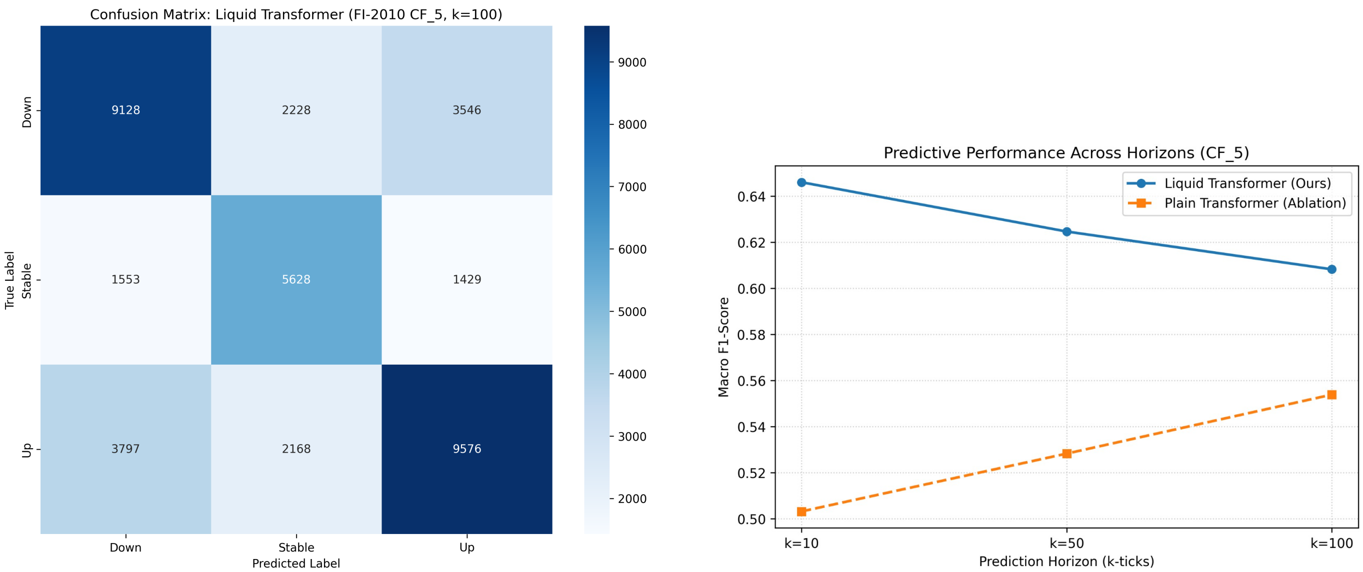
B. Discussion of Predictive Performance

Evaluated across three stock subsets (CF_1, CF_5, CF_9) of the NoAuction Z-Score normalized FI-2010 dataset, the Liquid Transformer achieves a mean Accuracy of 58.64% ± 6.21% and a mean Macro F1-Score of 0.5864 ± 0.0636. As shown in Table I, this outperforms both the MLP baseline ($F1 \approx 0.40$) and standard CNN-LOB models ($F1 \approx 0.53$) [2]. The per-subset variance (CF_1: $F1 = 0.4980$ to CF_9: $F1 = 0.6529$) reflects genuine stock-specific differences in order book microstructure, a known phenomenon in LOB research [1], and highlights the importance of multi-subset evaluation over single-subset reporting. Class-weighted loss was applied during training to address the inherent imbalance between Up, Down, and Stable classes. The remaining gap relative to TransLOB ($F1 = 0.6120$) is attributable to differences in training scope, as TransLOB aggregates results across all normalization schemes and stocks in FI-2010, whereas our evaluation isolates the Z-Score normalization scheme.

C. Ablation Study: Contribution of the Liquid Integration Head

To isolate the contribution of the Liquid Time-Constant (LTC) integration head, we compare the full Liquid Transformer against a Plain Transformer baseline — an identical architecture where the LTC head is replaced with a standard last-token readout ($x[:, -1, :]$). Both models are trained under identical conditions: weighted cross-entropy loss, Adam optimizer with $lr = 1 \times 10^{-4}$, and 20 epochs across three FI-2010 subsets (CF_1, CF_5, CF_9).

As shown in Table III, the LTC head consistently improves Macro F1-Score across all three subsets, with a mean improvement of $+0.060$ F1 points. The improvement is consistent across all three subsets, ranging from $+0.037$ on CF_1 to $+0.093$ on CF_9, indicating that the continuous-time hidden state provides a robust benefit across differing stock microstructures. These results confirm that the LTC integration head is the primary driver of performance gains over the plain Transformer baseline.



(a) Confusion Matrix at $k = 100$

(b) Performance across k (CF_5)

Fig. 1: Performance Analysis on the CF_5 subset: (a) confusion matrix at $k = 100$; (b) Macro F1-Score across horizons for the Liquid Transformer vs. the Plain Transformer ablation.

D. *Computational Efficiency and Throughput*

The operational cost of the Liquid integration head was evaluated via an inference latency benchmark on an NVIDIA T4 GPU, measuring wall-clock time over 1,000 iterations with batch size 1 to simulate real-time tick-by-tick processing. The naive LTC implementation, which evolves the hidden state sequentially across all 100 ticks, incurs a latency of 14.93 ms per tick. Applying TorchScript JIT compilation- without any change to the underlying computation- reduces this to 3.56 ms per tick, a $4\times$ speedup, corresponding to a throughput of approximately 280 ticks/s.

We report this latency transparently as a key finding: the continuous-time LTC head introduces a sequential ODE-solving dependency that cannot be parallelized across the time dimension, in direct contrast to the fully-parallel self-attention layers preceding it. This represents a fundamental *accuracy-latency tradeoff*: the LTC head improves Macro F1-Score by $+0.060$ (Table III) at the cost of increased per-tick latency. While 3.56 ms is sufficient for mid-frequency strategies, it exceeds the microsecond-scale budget of latency-critical HFT. We therefore identify hardware acceleration- specifically FPGA implementation of the ODE solver, which can parallelize the recurrence through pipelining- as the necessary path toward production HFT deployment.

E. *Error Distribution and Confusion Matrix Analysis*

The confusion matrix (Fig. 1(a)), computed on the CF_5 test set, shows the model correctly identifies 9,128 ”Down”, 5,628 ”Stable”, and 9,576 ”Up” movements. The relatively low frequency of inverse misclassifications (true ”Up” predicted as ”Down”, and vice versa) indicates that the ODE-based continuous hidden state maintains directional coherence across the prediction window.

F. *Performance Across Horizons*

As illustrated in Fig. 1(b), the Liquid Transformer improves Macro F1-Score over the Plain Transformer ablation at every horizon on the CF_5 subset. The advantage is largest at the shortest horizon ($+0.143$ at $k = 10$) and narrows as the horizon extends ($+0.054$ at $k = 100$), indicating that the continuous-time LTC head is especially effective at capturing the rapid, short-term order book dynamics that a discrete last-token readout fails to retain. While the Liquid Transformer's absolute F1 decays gently as the horizon lengthens, it maintains a consistent advantage over the plain baseline throughout.

VII. Conclusion

This research presented a Multi-Channel Transformer with Liquid Neural Integration for high-frequency LOB forecasting. By modeling hidden states as a continuous dynamical system, we demonstrated consistent predictive gains over a plain Transformer baseline across all tested horizons.

Experimental results on the FI-2010 dataset (k=100) highlight the model's effectiveness:

1) *Predictive Power:* Achieved a mean Accuracy of 58.64% $\pm$ 6.21% and a mean Macro F1-Score of 0.5864 $\pm$ 0.0636 across three FI-2010 subsets, outperforming CNN-LOB and plain Transformer baselines.

2) *Horizon Performance:* Demonstrated a consistent F1 improvement over the plain Transformer at every horizon, with the largest gains at short horizons.

3) Operational Speed and Tradeoff: Characterized a fundamental accuracy-latency tradeoff: the LTC head improves F1 by $+0.060$ but introduces a sequential dependency, yielding 3.56 ms per-tick latency after JIT optimization (a $4\times$ speedup over the naive implementation).

A. Limitations

Several limitations of this work should be acknowledged. First, evaluation is restricted to the NoAuction Z-Score normalization scheme of the FI-2010 dataset- results may differ under raw or decimal-precision normalization schemes. Second, the model is trained and tested independently per stock subset, with no cross-stock generalization evaluated; the observed variance across subsets (CF_1: $F1 = 0.4980$ to CF_9: $F1 = 0.6529$) suggests meaningful stock-specific dynamics that a single shared model may struggle to capture. Third, the inference latency reported reflects software-level GPU benchmarking rather than exchange co-located execution- production HFT deployment would require FPGA porting to achieve sub-microsecond latency.

These findings confirm that Liquid Neural Networks offer meaningful predictive gains for LOB forecasting, while highlighting that their sequential nature poses a latency challenge for the highest-frequency trading regimes. Future work will address three directions:

(1) extending evaluation to all nine FI-2010 subsets across multiple normalization schemes to improve generalizability;

(2) exploring cross-stock transfer learning to reduce per-subset performance variance; and

(3) porting the integration head to FPGA to achieve sub-microsecond latency for Tier-1 execution environments.

References

[1] A. Ntakaris, M. Magris, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods," *Journal of Forecasting*, vol. 37, no. 8, pp. 852–866, 2018.

[2] A. Tsantekidis, N. Passalis, A. Tefas, J. Kanniainen, M. Gabbouj, and A. Iosifidis, "Forecasting stock prices from the limit order book using convolutional neural networks," in *IEEE 19th Conference on Business Informatics (CBI)*. IEEE, 2017.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[4] R. Hasani, M. Lechner, A. Amini, D. Rus, and R. Grosse, "Liquid time-constant networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 9, 2021, pp. 7657–7666.